



Co-porphyrin-based metal-organic framework for light-driven efficient green conversion of CO₂ and epoxides

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ABSTRACT

The cyclization of CO₂ with epoxides to produce cyclic carbonate compounds represents a significant avenue for achieving CO₂ emission reduction and optimizing resource utilization. However, this process typically demands rigorous reaction conditions. In this study, we have successfully constructed three layered columnar porphyrin metal-organic frameworks (PMOFs) with double interpenetrated network configurations, denoted as Co-PMOFs 1–3, which have excellent photosensitivity, effective CO₂ adsorption, regularly ordered pore structures, and high-density monodispersed catalytically active centers by employing solvent-modulated and ligand-modulated construction strategies. Co-PMOFs 1–3 exhibit excellent catalytic performance in the traditional thermally-driven CO₂ cycloaddition reaction with epoxides under solvent-free conditions (75 °C and 1 bar). Notably, Co-PMOF 3 exhibits a wider pore environment, ensuring sustained efficient catalytic activity for larger-sized reaction substrates. Significantly, the incorporation of porphyrin and anthracene motifs enables the harnessing of solar energy, thereby enhancing the reaction efficiency under ambient temperature and pressure conditions. The majority of epoxides can be nearly completely transformed (> 99 %) into cyclic carbonates within a concise timeframe of 6 h. This investigation not only presents a distinguished class of MOFs-catalysts for CO₂ cyclization reactions but also imparts novel insights for the rational design of high-quality photocatalysts in future heterogeneous catalytic processes.

1. Introduction

With the rapid progress of industrialization, the combustion of fossil fuels has led to significant greenhouse gas emissions [1–6]. The excessive release of CO₂ has resulted in increasingly severe environmental challenges, encompassing global warming, rising sea levels, ocean acidification, and a heightened frequency of extreme weather events. These issues pose a substantial threat to the survival and progress of human society [2,7–13]. Consequently, the quest for sustainable, renewable, and energy-efficient methods to directly convert CO₂ into valuable chemical compounds has become a paramount objective for scientists and governments worldwide [14–16]. Among the various strategies, the cyclization reaction of CO₂ with epoxide to generate cyclic carbonates exhibits promise due to its 100 % atom economy. This approach not only diminishes the volume of carbon dioxide in the atmosphere but also transforms it into high-value fine chemicals [7,17–26]. However, due to the high stability and low reactivity of

carbon dioxide molecules, most catalytic systems for CO₂ cycloaddition require high temperatures, high pressures, or long reaction times, which do not align with the sustainable green chemistry concept [5,9,13,27]. As a result, the development of low-energy catalytic systems has become an imperative goal for researchers, aiming to achieve the environmentally friendly conversion of CO₂ with epoxides and enhance the overall sustainability of the conversion process.

Up to now, many excellent homogeneous and heterogeneous catalysts have been designed and constructed, which have been successfully applied in the cycloaddition reaction of CO₂ with epoxides. For instance, Li [26] and coworkers constructed an excellent single-atom catalyst, Ir₁-WO₃, by stabilizing Ir single atoms anchored to a WO₃ carrier. Ir₁-WO₃ exhibited exceptional catalytic activity in the cycloaddition reaction of CO₂ and epoxides: under solvent-free reaction conditions at 1 bar and 40 °C, it achieves 100 % conversion of styrene oxide to styrene carbonate within 15 h, and possesses excellent recyclability. In another instance, Chen [7] and coworkers dispersed Al atoms on porous carbon

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nanosheets and constructed Al-N-C catalysts by a multifunctional molecular confinement pyrolysis strategy. Al-N-C demonstrated efficient promotion of the CO₂ cycloaddition reaction under light irradiation (1.0 atm, DMF as the reaction solvent). Notably, small epoxides with diverse substituents were effectively converted to the desired products, although 2-(phenoxy)methyl)oxirane and styrene oxide exhibited comparatively lower reaction rates due to the pronounced steric hindrance from their larger substituents. These studies indicate that the cycloaddition reaction of CO₂ and epoxides often requires relatively harsh reaction conditions. However, from the perspective of green chemistry and sustainable development strategies, reaction conditions such as solvent-free, atmospheric pressure, and room temperature are more favored. If the efficient cyclization of CO₂ with epoxides can be driven by solar energy instead of thermal energy, it will be more in line with the development concept of green chemistry. In recent years, there have been many advancements in the photocatalytic synthesis of cyclic carbonates [28–30]. Photocatalytic synthesis can be carried out under mild conditions, such as room temperature and low CO₂ pressure, without the need for the high temperature and pressure required in traditional thermal catalysis, thereby reducing energy consumption and costs. Additionally, the synergistic effect between photocatalysis and photothermal catalysis can further enhance the reaction efficiency. Consequently, the development of efficient photocatalytic systems capable of promoting cyclization reactions between CO₂ and epoxides with varying sizes under mild conditions emerges as an exceptionally challenging yet indispensable endeavor.

Metal-organic frameworks (MOFs) [31–37] are an emerging class of crystalline porous materials, widely recognized for their ordered crystal structures, high surface area, and tunable structures. They hold extensive application prospects in various research fields, such as sensing, gas adsorption/separation, and catalysis [34,38–52]. Among them, MOFs have significant advantages over other types of catalysts in catalyzing the CO₂ cyclization reaction: (i) the diverse and modifiable structures of MOFs enable the incorporation of multiple catalytic sites, (ii) tunable pore structures and large specific surface areas in MOFs facilitate unhindered diffusion of substrates and products through the MOF channels, ensuring the proximity of active sites and effective mass transport, (iii) the high crystallinity of MOFs allows for the adsorption of CO₂ at

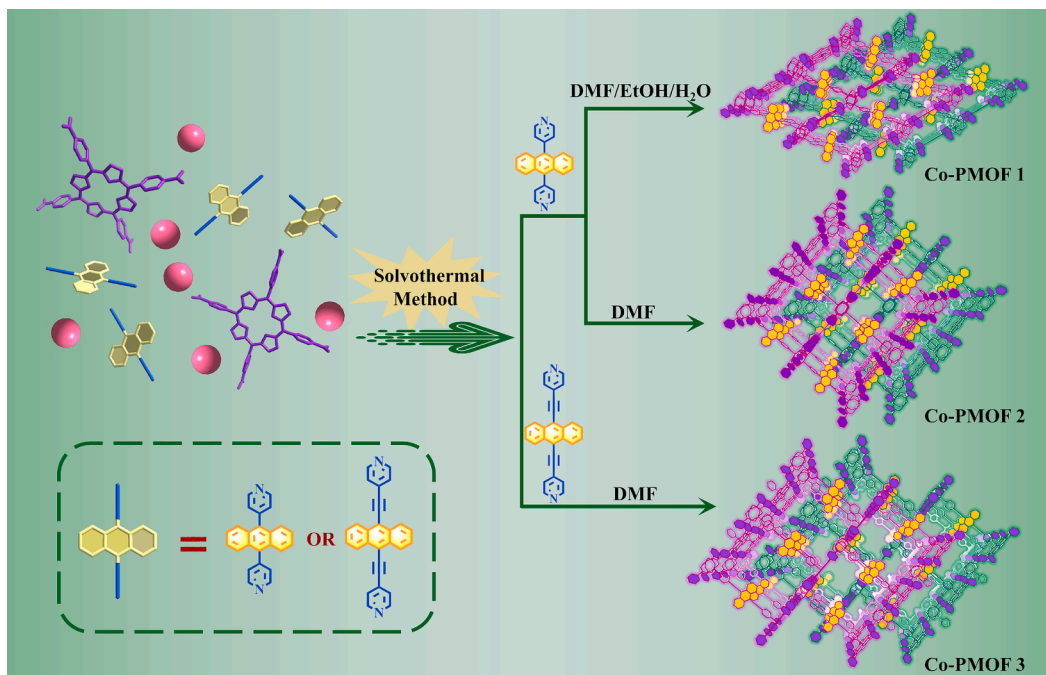
higher concentrations on the active sites, and (iv) the structural stability of MOFs facilitates their easy separation and recycling as heterogeneous catalysts [3,4,27,53–59]. Among various MOF catalytic systems for CO₂ cyclization reactions, porphyrin-based metal-organic frameworks (PMOFs) stand out for their exceptional potential in catalysis. This is attributed to their conjugated macrocyclic structures, tunable active sites, strong charge transfer ability, and enhanced thermal and photostability [60]. In the current research, PMOFs have been extensively employed in catalytic processes, particularly as photocatalysts for reactions such as CO₂ reduction, organic compound degradation or conversion, nitrogen fixation, and hydrogen/oxygen precipitation reactions for hydrolysis [55,61–73].

Based on the above considerations, we employed a hybrid ligand strategy to integrate photosensitive porphyrin and anthracene motifs into the framework of MOFs (Scheme 1). Using a simple solvothermal method, we successfully synthesized three double interpenetrated layered columnar Co-PMOFs. These prepared PMOFs have the following advantages: (i) a high specific surface area and photosensitive backbone, enhancing catalytic activity by promoting CO₂ enrichment and utilizing solar energy; (ii) tunable pore size and favorable adsorption affinity of the backbone for CO₂ molecules, facilitating substrate/product transport; (iii) atomically dispersed Co metal active sites that are readily accessible by epoxide and CO₂. Catalytic evaluations demonstrated that this series of photosensitive Co-PMOFs effectively catalyze the CO₂ cyclization reaction under mild and solvent-free conditions, exhibiting high efficiency and cyclic stability. Notably, Co-PMOF 3, characterized by a broader pore environment, maintains efficient catalytic activity for larger-sized reaction substrates. Furthermore, these catalysts demonstrate excellent stability. Additionally, detailed mechanistic pathways for the conversion of CO₂ into cyclic carbonates were elucidated through Density Functional Theory (DFT) calculations.

2. Experimental section

2.1. Synthesis of 4,4'-(9,10-anthracenediyl)bis-Pyridine (dpyaa) ligand

4-pyridylboronic acid (2.65 g, 21.56 mmol), 9,10-dibromoanthracene (3.02 g, 8.98 mmol), Cs₂CO₃ (5.85 g, 17.96 mmol) and Pd



Scheme 1. Schematic diagram of the synthetic routes for preparing Co-PMOFs 1–3.

(PPh₃)₂Cl₂ (0.38 g, 0.54 mmol) were added to a 200 mL Schlenk flask. The reaction vessel was evacuated under vacuum and flushed with N₂ for three times, then 80 mL 1,4-dioxane and 30 mL ultrapure water were injected. The reaction was reacted at 88 °C for 48 h. After cooling to room temperature, the mixture was extracted with dichloromethane. The combined organic layer was concentrated in vacuo, and then the crude product was purified by flash column chromatography with a solvent mixture of dichloromethane and ethanol ($V_{\text{dichloromethane}}/V_{\text{ethanol}} = 100/2$). The final yield was 55 %. ¹H NMR (400 MHz, CDCl₃, 300 K) δ (ppm) 8.93–8.86 (m, 4H), 7.65–7.59 (m, 4H), 7.47–7.43 (m, 4H), 7.43–7.37 (m, 4H). ¹³C NMR (100 MHz, CDCl₃, 300 K) δ (ppm): 150.47, 147.69, 135.06, 129.31, 126.76, 126.63, 126.27.

2.2. Synthesis of 4,4'-(9,10-anthracenediyl-di-2,1-ethynediyl)bis-Pyridine (bpyaa) ligand

Add 4-Iodopyridine (3.00 g, 14.63 mmol), Pd(PPh₃)₄ (0.68 g, 0.59 mmol) and CuI (0.11 g, 0.59 mmol) to 100 mL Schlenk flask. The reaction vessel was evacuated under vacuum and flushed with N₂ for three times. Then, Et₃N (15 mL), THF (50 mL) and Ethynyltrimethylsilane (4.31 g, 43.92 mmol) were added. The reaction mixture was then stirred at 75 °C for 12 h. After cooling to room temperature, the reactant was extracted with dichloromethane. Then evaporating the solvent under vacuum, the residue was purified by column chromatography on silica gel with dichloromethane as eluent to afford the compound 1 in 86 % yield. ¹H NMR (400 MHz, Chloroform-*d*) δ 8.65 (d, *J* = 5.2 Hz, 2H), 7.44–7.36 (m, 2H), 0.00 (s, 9H).

Compound 1 (2.00 g, 11.42 mmol) and K₂CO₃ (9.47 g, 68.54 mmol) were added to a 100 mL round bottom flask, and then 45 mL CH₃OH and 25 mL chloroform were poured into the reaction vessel. The mixture was stirred at room temperature for 4 h. After the reaction, it was extracted three times with dichloromethane and ultrapure water, then the organic layers were combined, and the organic solvent was removed under vacuum. Finally, the residue was purified by silica gel column chromatography (dichloromethane) to finally obtain compound 2 in 80 % yield. ¹H NMR (400 MHz, CDCl₃, 300 K) δ (ppm) 8.63–8.55 (m, 2H), 7.39–7.31 (m, 2H).

Under nitrogen, a mixture of compound 2 (0.81 g, 7.86 mmol), 9,10-dibromoanthracene (1.09 g, 3.28 mmol), CuI (0.12 g, 0.63 mmol) and Pd(PPh₃)₄ (0.92 g, 0.79 mmol) in 60 mL THF and 20 mL Et₃N was stirred at 70 °C for 12 h. After cooling to room temperature, the mixture was extracted with dichloromethane. After that, dichloromethane was removed and the residue was purified by column chromatography on SiO₂ with a solvent mixture of dichloromethane and ethanol ($V_{\text{dichloromethane}}/V_{\text{ethanol}} = 100/3$) as eluent to afford bpyaa in 60 % yield. ¹H NMR (400 MHz, CDCl₃, 300 K) δ (ppm) 8.75–8.71 (m, 4H), 8.65 (dd, *J* = 6.7, 3.2 Hz, 4H), 7.73–7.66 (m, 4H), 7.65–7.60 (m, 4H). ¹³C NMR (100 MHz, CDCl₃, 300 K) δ (ppm): 150.26, 132.59, 131.68, 127.81, 127.36, 125.86, 118.54, 100.02, 91.07.

2.3. Synthesis of Co-PMOF 1

Tetrakis(4-carboxyphenyl)porphyrin (TCPP, 0.005 mmol), dpyaa (0.01 mmol) and CoCl₂·6H₂O (0.015 mmol) were added to the solvent of 4 mL DMF/EtOH/H₂O (*v/v/v* = 3/1/1). This mixture was transferred into a 25 mL Teflon-lined stainless-steel vessel and reacted at 100 °C for 4 days. After gradually cooling to room temperature, purple crystals were obtained by filtration. These crystals were washed with ethanol and naturally air-dried. The yield was 35 % (based on TCPP).

2.4. Synthesis of Co-PMOF 2

The synthetic procedure of Co-PMOF 2 was the same as that of Co-PMOF 1 except that DMF/EtOH/H₂O (*v/v/v* = 3/1/1) was replaced by DMF. Purple lumpy crystals were acquired with a yield of 40 % (based on TCPP).

2.5. Synthesis of Co-PMOF 3

TCPP (0.005 mmol), bpyaa (0.01 mmol) and CoCl₂·6H₂O (0.015 mmol) were added to the solvent of 4 mL DMF. This mixture was transferred into a 25 mL Teflon-lined stainless-steel vessel and reacted at 100 °C for 4 days. After gradually cooling to room temperature, purple crystals were obtained by filtration. These crystals were washed with ethanol and naturally air-dried. The yield was 37 % (based on TCPP).

3. Result and discussion

3.1. Synthesis and characterizations of Co-PMOFs catalysts

3.1.1. Structural design of MOFs for CO₂ conversion

Owing to the distinctive versatility and facile structural adaptability of MOFs, it is feasible to create efficient MOFs catalysts by integrating specific structural motifs. In this study, we utilized photosensitive porphyrin/anthracene motifs and Lewis acid sites to successfully construct three 3D layered columnar Co-PMOFs. The synergistic coordination of porphyrin and anthracene organic ligands in MOF materials offers significant advantages, such as enhanced light absorption, improved separation efficiency of photogenerated charges, increased structural stability, and tunable electronic properties. These combined benefits are expected to significantly boost the photocatalytic performance of MOFs, leading to efficient and stable photocatalyst for CO₂ cyclization reactions.

The structure of MOFs could be regulated by adjusting the polarity of the reaction solvent or the length of the organic building blocks. Co-PMOFs 1–3 were 3D MOFs constructed in one step by solvothermal method, and the porphyrin centers were automatically metallized in the process of solvothermal method, forming Co-TCPP, which was then involved in the process of MOF structure construction. The single-crystal data showed that Co-PMOFs 1–3 belong to the *P* $\bar{1}$ space group of the triclinic crystal system, the *Cmmm* space group of the orthorhombic crystal system, and the *Imma* space group of the orthorhombic crystal system, respectively. The structures of Co-PMOF 1 and Co-PMOF 2 were similar and the difference was the structure of the binuclear due to the different polarity of the solvent used in the solvothermal reaction. In Co-PMOF 1, Co(II) was hexa-coordinated and the adjacent Co(II) was connected by four COO[−] in a bidentate bridging and bidentate chelating coordination pattern to form the [Co₂(COO)₄] binuclear cluster (Fig. S1). The distance between Co(II)1 and Co(II)1ⁱ was 2.8883 Å. Unlike in Co-PMOF 1, [Co₂(COO)₄] in Co-PMOF 2 displayed the shape of a Wheel and was connected by four COO[−] in a double-tooth bridging coordination mode with two Co(II). And the distance of Co(II)3 and Co(II)3^{vii} was 3.0274 Å. The binuclear cluster [Co₂(COO)₄] of Co-PMOF 3 showed a wheel-like shape, similar to that of Co-PMOF 2 (Fig. S6 and Fig. S11). As shown in Fig. 1, the main differences in the structure of Co-PMOFs 1–3 were the differences in the Co-TCPP coordination modes and the arrangement between neighboring 2D layers. We provided a detailed description of the single crystal structure of Co-PMOF 3, and the detailed structural descriptions of Co-PMOFs 1 and 2 were given in the [supporting information](#). Detailed structural comparisons of Co-PMOFs 1–3 were listed in [Table S10](#).

For Co-PMOF 3, it included three crystallographically independent Co(II) ions, 1/4 TCPP ligands, 1/2 bpyaa ligands, and two coordinated H₂O molecules in its asymmetric structural unit (Fig. S11). Co(II)1 was located in the center of the octahedral configuration and was coordinated in the equatorial plane to four pyrrole N atoms (N1ⁱ, N1^{vi}, N2ⁱ, N2ⁱⁱ) within the porphyrin ring, in axial position to one pyridine N atom (N27) as well as one H₂O (O2W) molecule in axial position. Co(II)2 was located at the center of the tetragonal conical conformation, coordinated to four carboxy oxygen atoms and one H₂O molecule (O1, O1ⁱⁱⁱ, O1^{iv}, O1^{vi}, O1W), respectively; Co(II)3 has a similar coordination pattern to Co(II)2, with four carboxyl oxygen atoms and one pyridine nitrogen

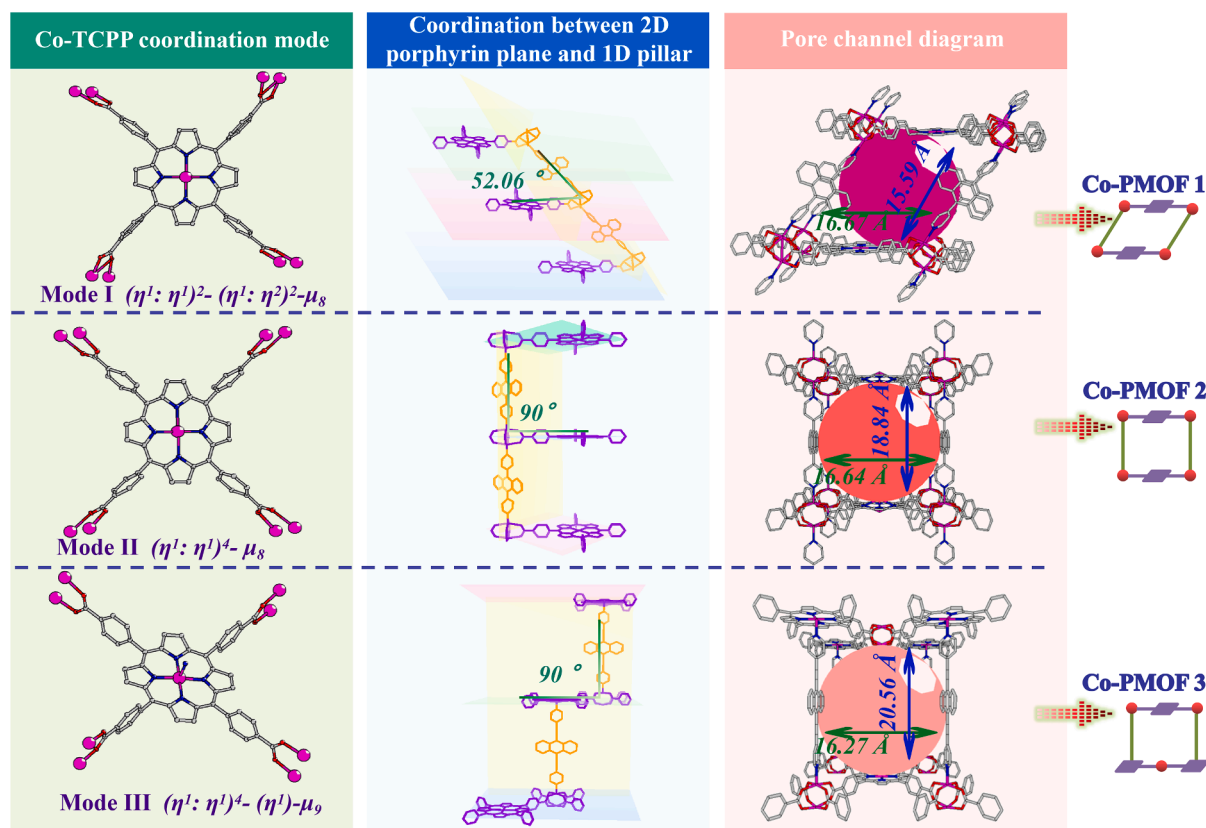


Fig. 1. Coordination modes of Co-PMOFs 1–3.

atom (O2, O2ⁱⁱⁱ, O2^{iv}, O2^v, N3). The bond lengths of Co–O and Co–N bonds were in the range of 1.9973 Å – 2.3101 Å and 1.9775 Å – 2.0781 Å, respectively, and all Co–O and Co–N bonds were in the reasonable bond length range. Co(II)3 and Co(II)2 were linked by four carboxylic acid groups in a ($\kappa^1-\kappa^1$)- μ_8 bidentate bridging mode to form a [Co₂(COO)₄(H₂O)N] wheel-like Secondary Building Unit (SBU).

As shown in the Fig. 2a, Co-PMOF 3 was a typical layered columnar 3D skeleton structure. The 2D planes were constructed by mutual coordination of TCPP ligands and [Co₂(COO)₄(H₂O)N] SBUs. However, Co-PMOF 3 showed significant differences from Co-PMOFs 1 and 2 in the connection way of between layers: adjacent layers were arranged in a staggered arrangement, with one end of the bpyaa connecting the binuclear SBU and the other end connecting the porphyrin-centered Co (II). The 2D planes were supported by bpyaa ligands with longer lengths to construct a 3D framework, giving Co-PMOF 3 a larger size 1D pore (20.56 Å × 16.27 Å) compared to Co-PMOFs 1 and 2. Co-PMOF 3 has regularly ordered 1D open pores along both the a-, b- and c-axis directions, which leads to the formation of a double interpenetrated 3D network structure (Fig. 2b). Despite the presence of interpenetrations, Co-PMOF 3 has pores with dimensions of 10.28 Å × 16.27 Å and 10.28 Å × 9.14 Å along the b-axis direction (Fig. 2c). The Co-PMOF 3 has 59 % void fraction (7053.3 Å³/12048.7 Å³) as obtained by PLATON calculations. Using ToposPro, the crystal structure of Co-PMOF 3 was topologically simplified. Both structural units, Co-TCPP and dinuclear SBU, can be simplified into 5-connected topological nodes. The framework of Co-PMOF 3 can be simplified as a three-dimensional double interpenetrated *sqc13* topological network, with a topological symbol of {4⁴.6⁶}. The interpenetrating network structure, with its three-dimensional interweaving, can significantly increase the surface area of the material, providing more channels for the adsorption and transport of reactants. Moreover, this structure enhances the stability of the framework, reducing the risk of catalyst deactivation due to framework collapse or structural changes during the reaction process.

The microscopic morphology of Co-PMOFs was investigated by using scanning electron microscopy (SEM) and transmission electron microscope (TEM). SEM and TEM images showed that Co-PMOFs 1–3 were similar in shape, all presenting a sub-square or rectangular shape (Figs. S15–S18, Fig. 3a and b). And the energy-dispersive spectrometry (EDS) elemental mapping analysis showed that C, N, O, Co are uniformly distributed in the PMOFs. As shown in Fig. 3c, thermogravimetric analysis (TGA) showed that Co-PMOFs 1–3 have excellent thermal stability, maintaining the integrity of the skeleton structure until 415 °C, 440 °C and 415 °C, respectively. The phase purity of Co-PMOFs 1–3 was verified by powder X-ray diffraction (PXRD) test. From Fig. S27, Fig. S28 and Fig. 3d, it can be seen that the experimental powder diffraction patterns of the PMOFs samples basically match with their corresponding simulated diffraction patterns, indicating that the obtained PMOFs samples have good phase purity and can be used for further testing and characterization.

3.1.2. Gas adsorption studies

The porosity of Co-PMOFs was verified by N₂ adsorption experiment. Activation of PMOFs samples was required prior to adsorption testing. First, the newly prepared samples were immersed in methanol solvent for sufficient solvent exchange (3 days), and then dried under vacuum at 130 °C for 12 h to remove the lattice solvent to obtain the corresponding activated PMOFs samples. Concurrently, the structural integrity of the activated Co-PMOFs 1–3 was confirmed by PXRD shown in Figs. S27, Fig. S28 and Fig. 3d. The inherent permanent porosity of Co-PMOFs 1–3 was verified by the reversible adsorption of N₂ at 77 K. As can be seen from Fig. 3e, the N₂ adsorption curves of Co-PMOFs 1–3 are all typical type I adsorption–desorption isotherms with Brunauer–Emmett–Teller (BET) surface areas of 188.23 m²·g^{−1}, 373.19 m²·g^{−1} and 579.72 m²·g^{−1}, respectively. From the adsorption–desorption curves of N₂, a sharp increase in the adsorption isotherm in the low-pressure region indicates the microporous properties with PMOFs. Additionally, the

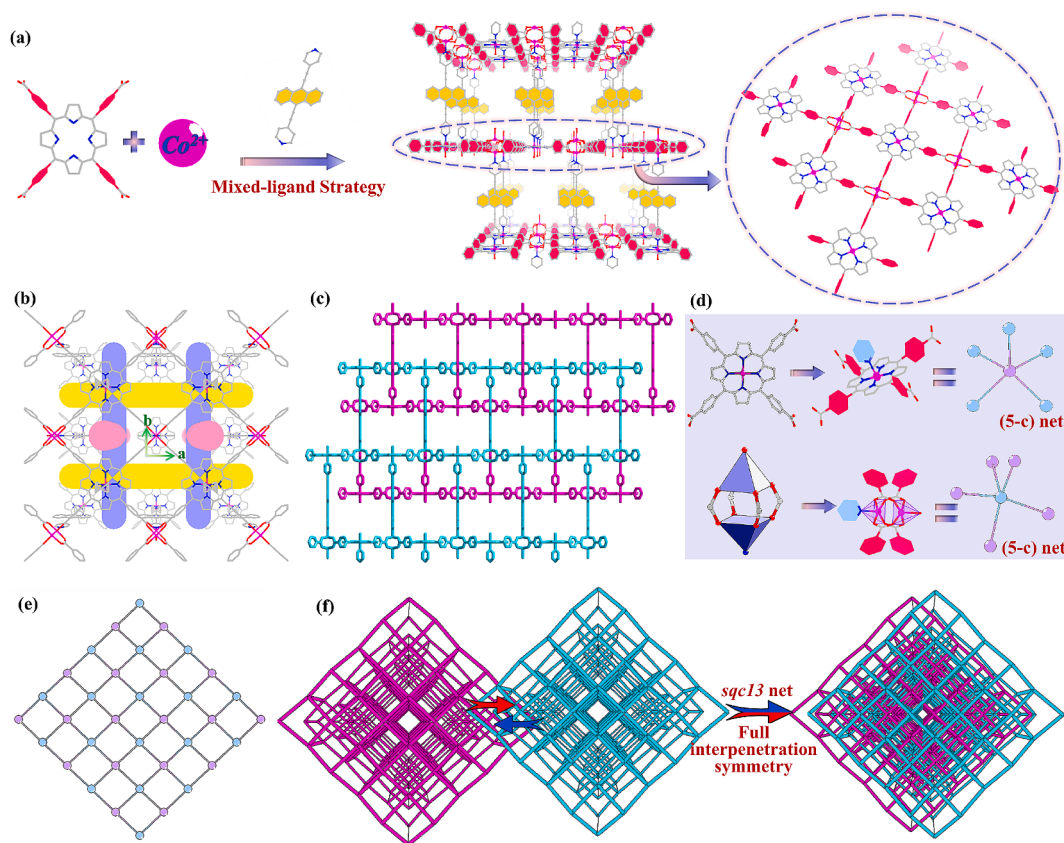


Fig. 2. Single-crystal structure of Co-PMOF **3**. (a) Three-dimensional framework along the a-axis, (b) three-dimensional framework along the c-axis, (c) double interpenetrated structure diagram, (d-f) the topological structure. $\text{I}^{-1/2} + \text{X}$, $-1/2 + \text{Y}$, $-1/2 + \text{Z}$; $\text{II}^{3/2} - \text{X}$, $1 - \text{Y}$, $-1/2 + \text{Z}$; $\text{III} + \text{X}$, $1/2 - \text{Y}$, $+ \text{Z}$; $\text{IV}^{1} - \text{X}$, $+ \text{Y}$, $+ \text{Z}$; $\text{V}^{1} - \text{X}$, $1/2 - \text{Y}$, $+ \text{Z}$; $\text{VI}^{2} - \text{X}$, $+ \text{Y}$, $+ \text{Z}$; $\text{VII} + \text{X}$, $3/2 - \text{Y}$, $+ \text{Z}$.

pore size distribution analyzed through density functional theory (DFT) calculations indicates that Co-PMOF **3** has the widest and most uniform pore structure compared to Co-PMOF **1** and Co-PMOF **2**, which facilitates the transport and diffusion of substrates within the pores (Fig. S29). The adsorption performance of the PMOFs material on CO_2 was further investigated. Given the larger pore structure and higher BET specific surface area of Co-PMOF **3**, it has a higher CO_2 adsorption capacity relative to Co-PMOF **1** and Co-PMOF **2**, with a maximum capacity of $39.1 \text{ cm}^3 \cdot \text{g}^{-1}$ and $26.7 \text{ cm}^3 \cdot \text{g}^{-1}$ at 273 and 298 K, respectively (Fig. 3f and g). In addition, the enthalpy of adsorption (Q_{st}) was calculated based on the Verrill method to determine the binding affinity between the PMOFs framework and the CO_2 molecules. The Q_{st} values of Co-PMOFs **1–3** at zero loading were classified as $23.6 \text{ kJ} \cdot \text{mol}^{-1}$, $18.2 \text{ kJ} \cdot \text{mol}^{-1}$ and $26.2 \text{ kJ} \cdot \text{mol}^{-1}$ (Fig. 3h), respectively, indicating a good adsorption affinity between the materials and CO_2 , thus favoring the enrichment of CO_2 molecules.

3.2. Solvent-free fixation of CO_2 conversion

From these structural and performance characterizations, it can be seen that the layered columnar Co-PMOFs **1–3** with double interdigitated topological configurations have good pore structures, atomically dispersed unsaturated Lewis metal acidic sites, excellent thermodynamic stability, and good adsorption affinity for CO_2 molecules. The combination of the above properties made Co-PMOFs **1–3** as heterogeneous catalysts have the potential to catalyze the conversion of CO_2 . Therefore, we evaluated the performance of this series of materials as heterogeneous catalysts for CO_2 and epoxide cycloadditions.

3.2.1. Thermocatalytic fixation of CO_2

We first evaluated the material's thermally driven CO_2 cycloaddition

properties with epoxide. The catalytic reaction was carried out in a Schlenk tube by utilizing a solid–liquid heterogeneous reaction system. In order to optimize the reaction conditions, epichlorohydrin (ECH) was chosen as the reaction substrate to explore the optimal reaction conditions of Co-PMOF **1** as heterogeneous catalyst for CO_2 cycloaddition reaction. Initial catalytic studies found that at ambient temperature and pressure (1 bar CO_2 and 25°C with tetrabutylammonium bromide (TBAB) as cocatalyst), the conversion of bromopropylene oxide was about 23.7 % in 10 h. No significant increase in conversion was observed when the reaction time was extended to 12 h. When we increased the reaction temperature to 45°C and 65°C , the conversion of ECH was 56.5 % and 73.7 % in 10 h, respectively (Fig. 4a). When the reaction temperature reaches 75°C , ECH was gradually converted to cyclic chloropropenyl carbonate with the increase of reaction time. The reaction substrate was almost completely converted with 98.4 % conversion within 10 h of reaction time as shown in Fig. 4c and d. Furthermore, as demonstrated by the control experiments (Fig. 4b), the presence of the MOF catalyst, TBAB co-catalyst, and an appropriate reaction temperature are all crucial for the successful progression of the reaction. Each of these components is essential, and the reaction cannot proceed effectively if any one of them is absent. Therefore, the reaction conditions for the following tests were determined as 0.1 mol% Co-PMOFs and 5 mol% TBAB co-catalyst, a reaction temperature of 75°C , a pressure of 1 bar, and a reaction time of 10 h. Compared with many MOFs-based catalysts reported in the literature, such as PCP-33(Mn) [74], Ni-TCPE1 [75], PCN-224(Co) [55], etc., which usually require high temperatures, high pressures, or long reaction times in order to drive the reaction, the use of Co-PMOF **1** as a catalyst for such reactions allows the reaction to proceed under relatively mild conditions.

Similarly, with Co-PMOF **2** or Co-PMOF **3** as catalysts, ECH can be almost completely converted to cyclopropylene carbonate within 10 h

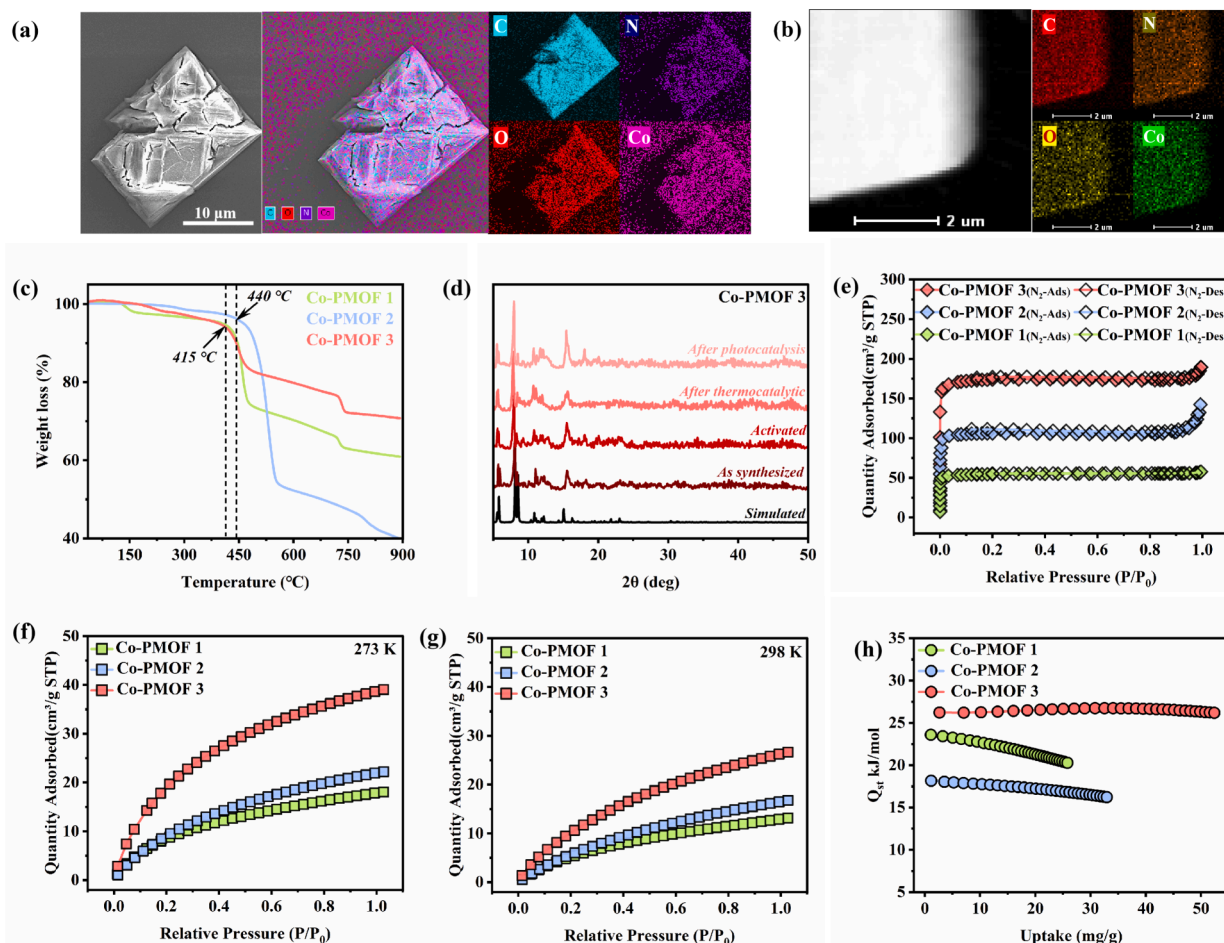


Fig. 3. (a) SEM images and EDS of the Co-PMOF 3, (b) TEM images and EDS of the Co-PMOF 3, (c) TGA curves of Co-PMOFs 1-3, (d) XRD of Co-PMOF 3 (e) N_2 adsorption isotherms of Co-PMOFs 1-3 at 77 K, (f and g) CO_2 adsorption/desorption isotherms analysis at 273 K and 298 K for Co-PMOFs 1-3, (h) corresponding Q_{st} values for Co-PMOFs 1-3.

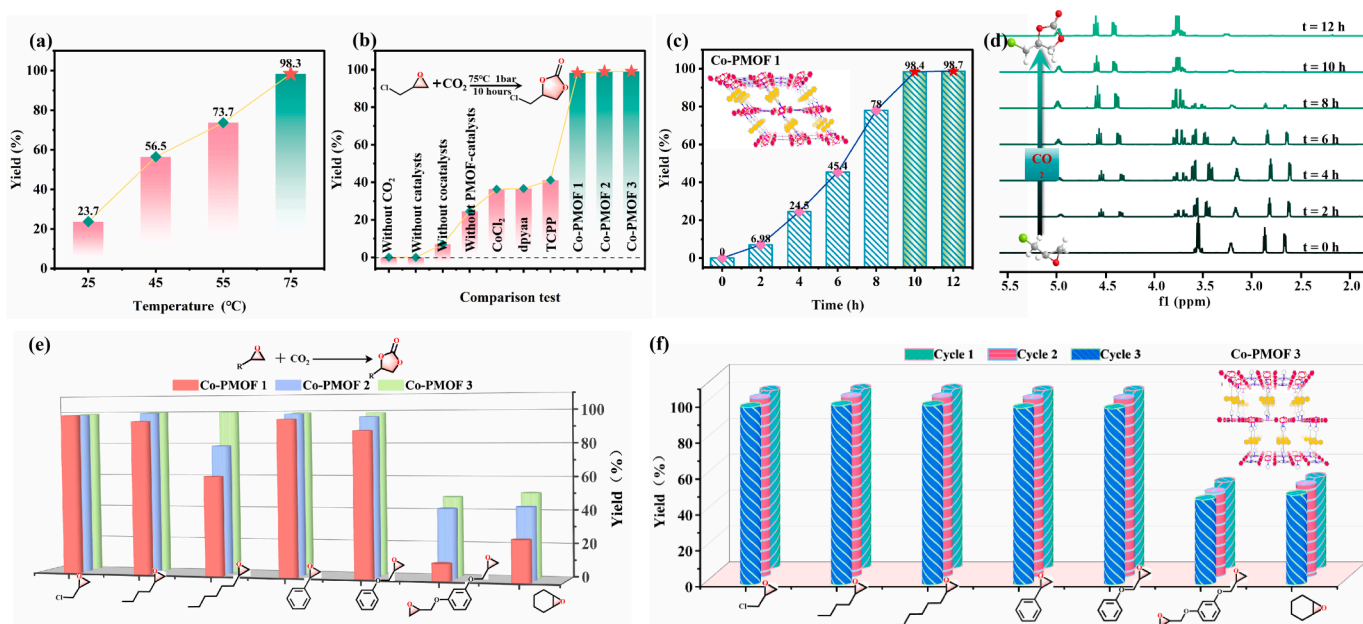


Fig. 4. Thermocatalytic fixation of CO_2 by Co-PMOFs: (a) Cyclization yields of CO_2 with epichlorohydrin at different temperatures, (b) yields of the control experiment, (c and d) catalytic performance of Co-PMOF 1 (reaction yield versus time), (e) comparison of catalytic performance of Co-PMOFs 1-3 with different epoxides for CO_2 cycloaddition reactions, (f) cyclic stability of Co-PMOF 3.

(conversion > 99 %). Encouraged by the high catalytic activity of this series of PMOFs materials, we also investigated the catalytic activity of the Co-PMOFs 1–3 catalysts for various other epoxides including oxidized styrene, 1,2-epoxyhexane, 1,2-epoxyoctane, phenyl glycidyl ether, resorcinol diglycidyl ether, and cyclohexene oxide, under optimized conditions, to validate their applicability for the catalytic CO₂ and epoxide cycloaddition reactions. As shown in Fig. 4e, Co-PMOFs catalysts maintained their catalytic efficacy for a wide range of aromatic and aliphatic substrates. Additionally, the results indicated that the smaller epoxides were nearly entirely converted to the corresponding cyclic carbonates, while for the larger epoxides, the conversion decreased with increasing epoxide size due to the larger epoxides showed limited diffusion in the narrow voids of the microporous framework. Co-PMOFs 1 and 2 with relatively small pore sizes showed lower catalytic activity for resorcinol diglycidyl ether, 10.7 % for Co-PMOF 1 and 42.9 % for Co-PMOF 2. On the contrary, Co-PMOF 3 with large pore sizes showed the improved catalytic activity (47.4 %) for resorcinol diglycidyl ether under the same reaction conditions. The conversion rate of cyclohexene oxide as substrate could reach 25.4 %, 43.5 % and 50.7 % for Co-PMOF 1, Co-PMOF 2 and Co-PMOF 3, respectively. The wider pore environment of Co-PMOF 3 facilitates the transport and diffusion of substrates within the catalyst's pores, enhancing the interaction between reactants

and catalytic sites. The higher CO₂ adsorption affinity allows the catalyst surface to accumulate more CO₂ molecules, effectively increasing the concentration of reactants near the active sites. Additionally, the larger surface area provides more active sites, enabling substrates to make more efficient contact with the catalytic centers. These three factors work synergistically, resulting in higher catalytic activity in the CO₂ cycloaddition fixation process.

3.2.2. Photocatalytic fixation of CO₂

It is noteworthy that reaction conditions such as solvent-free, atmospheric pressure and room temperature were highly desirable considering the principles of green chemistry. Inspired by photocatalysis, the photogenerated electrons generated on the semiconductor can activate CO₂. Considering that the photoactivated CO₂ has higher activity than neutral CO₂ under mild reaction conditions, it was believed that the photogenerated electrons can improve the efficiency of catalytic reaction. Presently, there were few reports on the introduction of light into the cyclization reaction of epoxides with CO₂ relative to thermal catalysis. Therefore, the direct use of solar energy instead of thermal energy to drive the cyclization reaction of CO₂ with cyclic carbonates will be more responsive to the required energy and environmental requirements, and conform to the new concepts of green chemistry and

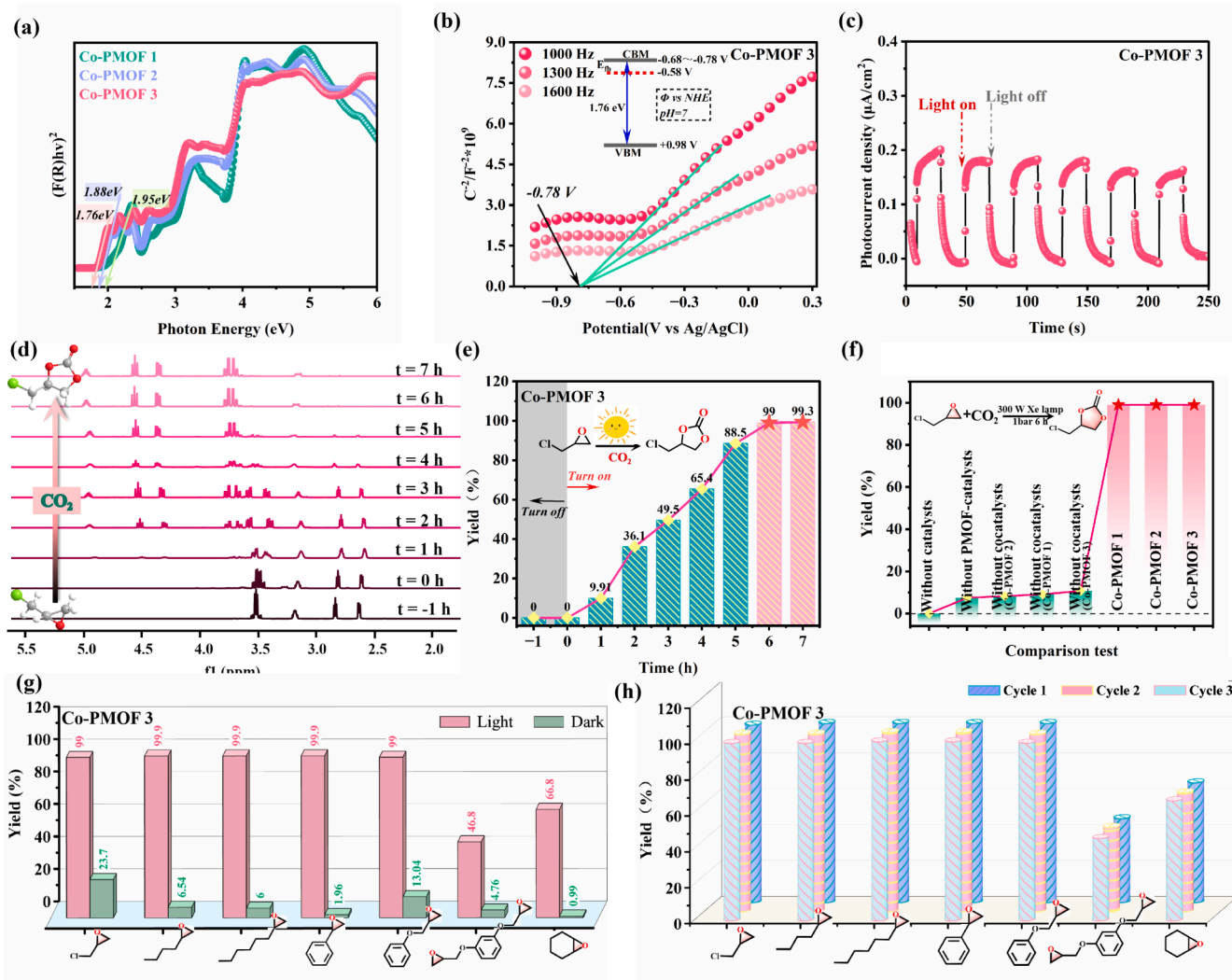


Fig. 5. Photocatalytic fixation of CO₂ by Co-PMOFs: (a) diffuse reflectance spectrum of Co-PMOFs 1–3 via K-M transformation, (b) Mott-Schottky plot of an electrode containing Co-PMOF 3 and calculated redox potentials, (c) photocurrent-time (i-t) curves of Co-PMOF 3, (d and e) catalytic performance of Co-PMOF 1 (reaction yield versus time), (f) yields of the control experiment, (g) comparison of catalytic performance of Co-PMOF 3 for CO₂ cycloaddition reactions with different epoxides under light and dark environments, respectively, (h) cyclic stability of Co-PMOF 3.

sustainable development. Given the introduction of highly photosensitive porphyrin and anthracene moieties in this series of PMOFs, this prompted us to investigate their application in light-driven CO₂ fixation.

In order to evaluate the light trapping ability of this series of PMOFs materials, ultraviolet–visible diffuse reflectance spectroscopy (UV–vis DRS) analysis was carried out. From the UV–Vis DRS spectra, it can be seen that the Co-PMOFs 1–3 catalysts exhibited a wide spectral absorption range, showing strong absorption peaks from 200 nm to 800 nm, which almost covers the ultraviolet and visible regions of sunlight region (Fig. S19). This implied their potential application in efficient solar energy harvesting to enhance the activity of light-driven CO₂ cyclization reactions. The bandgap energies (E_g) of Co-PMOFs 1–3 were estimated to be 1.95 eV, 1.88 eV, and 1.76 eV, respectively, based on the Tauc function, which were very close to the bandgap values of other PMOFs (Fig. 5a) [76–78]. The calculated band gap values highlight the visible light absorption properties of this series of PMOFs materials. Among them, Co-PMOF 3 has the smallest bandgap, indicating stronger light absorption capability in the visible light range. In view of the efficient thermal catalytic activity and broader substrate selectivity of Co-PMOF 3, we used 3 as an example to further explore the catalytic activity of this photosensitive PMOF for the light-driven fixation of CO₂. First, Mott-Schottky analyses were performed at different frequencies of 1000 Hz, 1300 Hz, and 1600 Hz to evaluate the sideband positions of Co-PMOF 3. As shown in Fig. 5b, the positive slope of the curve indicated that Co-PMOF 3 was n-type with an E_{fb} value of -0.78 V compared to the Ag/AgCl electrode (-0.58 V compared to the NHE at pH 7). Since the E_{fb} of the n-type semiconductor was considered to be about 0.1 V -0.2 V below the CBM, the CBM of Co-PMOF 3 was estimated to be -0.68 V to -0.78 V (NHE). Considering the bandgap value, the VBM of Co-PMOF 3 was calculated to be $+0.98$ to $+1.08$ V vs NHE. The photocurrent response and electrochemical impedance spectroscopy (EIS) were used to explore the photo-charge separation efficiency of Co-PMOF 3. The photocurrent response was performed in a standard three-electrode system with the Co-PMOF 3 sample modified ITO substrate as the working electrode. As shown in Fig. 5c, the transient current density-time curves indicated that the photocurrent density of Co-PMOF 3 under simulated sunlight irradiation increased dramatically due to the easy separation of carriers and decays to zero when the light was turned off. This was due to the rapid separation and migration of carriers to the reaction sites, and thus Co-PMOF 3 showed an effective photo-responsivity. In addition, EIS measurements showed that Co-PMOF 3 has a lower charge transfer resistance, which leads to more efficient charge separation and less charge transfer resistance under light irradiation (Fig. S20). Photo-induced charge separation can greatly improve the catalytic performance of the material in light-driven CO₂ cycloaddition.

In view of broader substrate selectivity of Co-PMOF 3, we used it as an example to further explore the catalytic activity of the light-driven fixation of CO₂ of this photosensitive MOFs. The reaction system was exposed to a 300 W xenon arc lamp (PeffetLight, PLS-SXE300C) providing full-spectrum light. To eliminate the thermal effects caused by light irradiation, we used a cooling water circulation system to maintain the reaction temperature at 25 °C. The dosages of substrate, catalysts, and cocatalysts used in the photocatalytic process are identical to those used in the thermal catalytic process. Under the conditions of RT and 1 bar CO₂, using Co-PMOF 3 as the catalyst and TBAB as the cocatalyst, the conversion of ECH gradually increased with increasing reaction time under simulated sunlight irradiation, and ECH was almost completely converted to CC with 99 % conversion within 6 h (Fig. 5d and e). The conversion rate was 1.67 times that of the thermal catalytic conversion rate. However, under dark conditions, the conversion of ECH was only 23.7 % within 6 h and showed almost no increasing trend with the increase of reaction time. These results suggested that under mild conditions (RT and 1 bar CO₂), light promotes the acceleration of the catalytic activity of the conversion of ECH to CC and was a key factor driving the proper performance of this catalytic reaction. Furthermore,

when using only Co-PMOF 3 or only TBAB as the reaction catalyst, the conversion rates of epichlorohydrin were 10.7 % and 7.41 %, respectively (Fig. 5f). Therefore, in the photocatalytic CO₂ fixation reaction, the presence of sunlight, the catalyst, and the cocatalyst is indispensable for achieving high efficiency. Encouraged by the high photocatalytic activity of Co-PMOF 3, the catalysis was extended for other epoxides (both aliphatic and aromatic), and the results were shown in Fig. 5g. Noticeably, most of the epoxides were converted to their respective CCs with high yield under the mild conditions of RT and 1 bar conditions under simulated sunlight irradiation. The conversion of resorcinol diglycidyl ether and cyclohexene oxide could also reach 46.8 % and 66.8 %.

3.2.3. Recyclability of materials

Under the optimized reaction conditions, the cycling stability of Co-PMOFs catalysts was confirmed by recyclability experiments. Taking Co-PMOF 3 with the best catalytic conversion performance as an example, the catalyst was separated by centrifugation after the completion of the reaction, washed three times repeatedly with ethanol and dichloromethane, and dried in a vacuum drying oven at 130 °C before being used in the next catalytic experiment. As can be seen from Figs. 4f and 5h, the Co-PMOF 3 catalysts maintained good catalytic performance in the reaction with almost no change in the catalytic activity after three cycles of cycling experiments, both in thermal catalysis and photocatalysis. Analysis of the recovered Co-PMOFs samples using IR, PXRD, SEM, TEM, ICP-OES and ¹H NMR indicated that Co-PMOFs maintained their original framework integrity throughout the reaction, demonstrating excellent structural stability (Figs. S24–S28, Figs. S30–S38, Fig. 3d, Supporting Information Table S11). The above experimental results showed that this series of Co-PMOFs can serve as excellent heterogeneous catalysts, which can effectively promote the reaction of CO₂ with epoxides to form cyclic carbonates under mild and solvent-free conditions. Additionally, compared to most catalysts reported in the literatures [7,26,30,63,64], this series of materials demonstrates broader substrate applicability, operates under milder reaction conditions, and exhibits higher catalytic efficiency (Supporting Information Table S12). Compared to traditional metal porphyrin catalysts, Co-PMOFs is a three-dimensional doubly interpenetrated network structure with offers higher surface area, denser catalytic active sites, and greater structural stability, making them more efficient for the adsorption and conversion of CO₂ and epoxides. Additionally, as heterogeneous catalysts, Co-PMOFs are easy to recover and reuse, exhibit low metal leaching, and maintain catalytic activity over multiple cycles. This combination of high stability, reusability, and efficient catalytic activity significantly enhances the innovative potential of Co-PMOFs in photocatalytic applications.

3.3. Exploration of the catalytic mechanism toward CO₂ cycloaddition

We visually explored the activation capacity of Co-PMOF 3 for CO₂ by in situ FTIR spectroscopy of CO₂ adsorption. The in situ FTIR spectrum of adsorbed CO₂ on Co-PMOF 3 reveals a ν_3 asymmetric stretching band of CO₂ at 2359 cm⁻¹ as well as four larger shoulder peaks (2329 cm⁻¹, 2331 cm⁻¹, 2336 cm⁻¹ and 2339 cm⁻¹) (Fig. S39). The above peaks on Co-PMOF 3 were still visible after 15 min of argon purge, a phenomenon that suggests a strong interaction between CO₂ and the corresponding sites. The blue shift to 2359 cm⁻¹ compared to the ν_3 mode (2349 cm⁻¹) of CO₂ in the gas phase was attributed to the electrostatic interaction between the open Co(II) sites and CO₂. The DFT-calculated IR spectrum (Fig. 6a) confirmed that the five shoulder peaks are attributed to the O = C = O bending vibrations at the adsorption sites. It can be seen that the bond angles of O = C = O bend from 180° to 178.1°, 177.7°, 179.3°, 177.9°, and 179.0° in configurations 1–5, confirming that CO₂ was activated on Co-PMOF 3.

In order to analyze the reaction mechanism involved in the transformation process, we have illustrated the key intermediates and

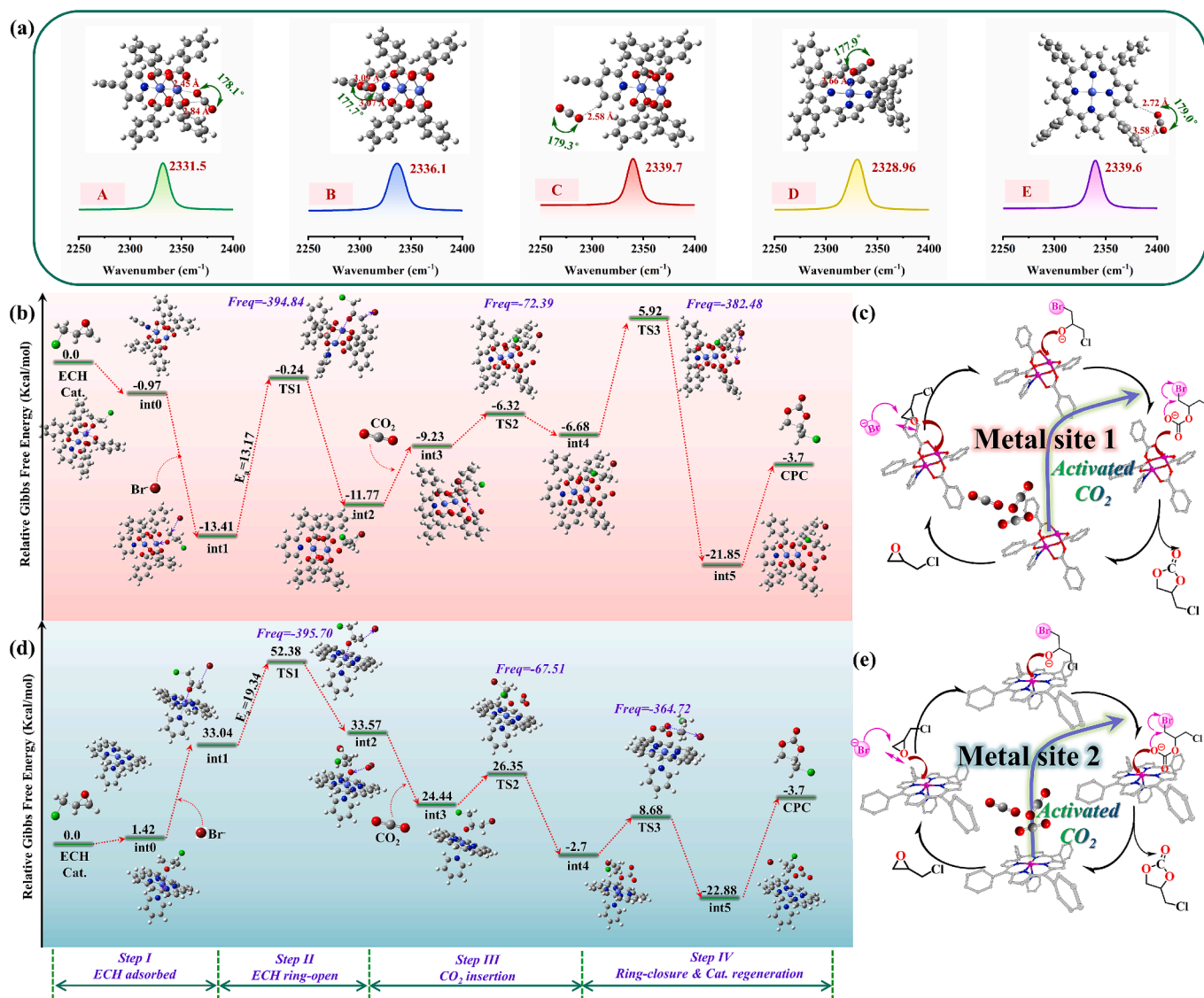


Fig. 6. Catalytic mechanism of Co-PMOF 3: (a) DFT calculated IR for spectra and corresponding models of Co-PMOFs adsorbed with CO₂, (b and d) the relative free energy profiles of the intermediates and transition states of the cycloaddition of ECH and CO₂ catalyzed by Co-PMOF 3, (c and e) schematic of the mechanism for the coupling reaction of CO₂ with epoxides catalyzed by Co-PMOF 3.

reaction routes in the catalytic cycloaddition of epoxides (e.g., ECH) with CO₂ in the presence of Co-PMOF 3 and TBAB by DFT calculations. Two types of active sites were present in Co-PMOF 3, the Co(II) Lewis acidic site (site 1) in the binuclear SBUs and the porphyrin-centered ligand-unsaturated Co(II) Lewis acidic site (site 2). Here, TBAB provided a nucleophile (Br⁻) to the pore of Co-PMOF 3, and the epoxide (in the case of ECH) ligated to the Co(II) metal catalytic site while CO₂ molecules were enriched in the Co-PMOF 3 pore, thus driving the catalytic reaction. As shown in Fig S40, it was the non-catalytic reaction mechanism without the introduction of PMOF catalyst in the reaction system. The ring-opening process, as the absolute rate step of the reaction, has an activation potential barrier of 70.5 kcal·mol⁻¹. The high activation energy barrier indicated that the ring-opening process is very difficult to carry out, which demonstrates the importance of the introduction of the catalyst. When Co-PMOF 3 was introduced as a catalyst in the reaction system, both site 1 and site 2 can act as catalytic activity centers for the CO₂ cyclization reaction. The relative potential energy curves as well as the optimized structures of the intermediates and transition states when the two metal centers were used as the active centers of the reaction, respectively, were provided in Fig. 6b and d. The initial stage of the reaction involved the ring opening of ECH, where the

oxygen atom O(ECH) of ECH interacts with the Co(II) metal center to form the intermediate (int0); the Br⁻ ion nucleophilic attacked on the less sterically hindered carbon atoms on the ECH ring forms the intermediate (int1); and the ring-opening reaction occurred to produce the first transition state (TS1). The relative energy difference between TS1 and int1 for site 1 and site 2 was 13.17 kcal·mol⁻¹ and 19.34 kcal·mol⁻¹, respectively, and it can be seen that the introduction of the Co-PMOF 3 catalyst significantly reduced the decisive step energy barrier for the fixation of CO₂ with ECH. Site 1 and site 2 synergize to improve the conversion efficiency of the reaction. Next, as the cyclization reaction proceeded, CO₂ molecules were inserted into the brominated alkanes to form the second transition state (TS2), which was formed by the strong interaction of CO₂ with the negatively charged O(ECH), and the relative energy difference between the catalytically active sites of TS2 and int3 was 2.91 kcal·mol⁻¹ and 1.91 kcal·mol⁻¹, respectively. In the final stage of ring closure, the intramolecular closure is completed by passing through the third transition state (TS3) to form cyclic carbonates.

Based on previous reports and the results obtained in the present study, the synergistic catalytic mechanism of cycloaddition reaction was proposed in Fig. 6c and e (Co-PMOF 3 as an example). Firstly, the Lewis acidic center of activated Co-PMOF 3 firstly participated in the

coordination of the epoxide substrate. Then, the nucleophilic bromide ions in the co-catalyst make a nucleophilic attack on the epoxide, leading to ring opening. The O-atoms of the ring-opened epoxide then react with CO₂. Finally, intramolecular ring closure occurred, resulting in the formation of cyclic carbonate and the regeneration of the catalyst. The synergistic effect of the two active metal centers in Co-PMOF **3** promoted the CO₂ cycloaddition reaction. Specifically, in the process of photocatalytic CO₂ cycloaddition, Co-PMOF **3** undergo charge separation upon light excitation, where photogenerated electrons in the VB migrate to the CB. The high-energy electrons accumulated in the CB effectively activate CO₂, converting it into the more reactive CO₂^{•-} radical anion. Meanwhile, the photogenerated holes in the VB act as Lewis acids, promoting the ring-opening of epoxides and generating the corresponding radical cations. Therefore, compared to thermally driven processes, Co-PMOF **3** exhibits greater reactivity in photocatalytic CO₂ cycloaddition.

4. Conclusion

In summary, we successfully synthesized three layered columnar porphyrin-based metal-organic frameworks with double interdigitated network topological configurations. This achievement was realized through the incorporation of photosensitive porphyrin and anthracene motifs into the MOF structures, employing a hybrid ligand strategy. This series of Co-PMOFs can be used as heterogeneous catalysts to efficiently drive the green cyclization of a wide range of aliphatic and aromatic epoxides and CO₂ under mild and solvent-free reaction conditions. The distinctive structural features of Co-PMOF **3**, characterized by cooperative interactions between active metal centers and a larger pore size, enhance its substrate universality, maintaining high catalytic activity for larger-sized reaction substrates. Notably, unlike traditional catalyst materials, the exceptional photosensitivity of this material series enables efficient utilization of sunlight to orchestrate the cyclization of CO₂ at ambient temperature and pressure. Under simulated sunlight irradiation, Co-PMOF **3** exhibited remarkable catalytic performance, ECH was almost completely converted to CC with 99 % conversion rate in 6 h, which was 1.67 times higher than the rate of thermocatalytic conversion. In addition, the detailed reaction mechanism of PMOFs catalyst materials involved in the cycloaddition reaction of CO₂ was investigated by DFT calculations. The novel PMOFs heterogeneous catalytic system developed in this study provides innovative concepts for the rational design of excellent catalysts in future CO₂ cycloaddition reactions.

CCDC 2327308, 2327309 and 2327311 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via http://www.ccdc.cam.ac.uk/data_request/cif.

CRediT authorship contribution statement

Jinxia Liang: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Xin Jiang:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Conceptualization. **Xinrui Zhang:** Writing – original draft, Visualization, Validation, Software, Data curation. **Hao Yu:** Writing – original draft, Software, Data curation. **Junjun Shi:** Writing – review & editing, Writing – original draft, Supervision, Formal analysis. **Ming Wang:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Ming Wang reports financial support was provided by National Natural Science Foundation of China. If there are other authors, they declare that

they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ccej.2024.156428>.

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